

EXPLORATORY DATA ANALYSIS PROJECTS

EDA01: App Insights Analysis
EDA02: Hotel Operations Analysis
EDA03: Heart Health Analysis

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Academic Project Report

Declaration and Note

This document provides a complete project structure and analysis plan for all three projects listed in the supplied checklist. The checklist identifies the Google Play Store, hotel booking, and heart health topics, but it does not include the underlying datasets or detailed resource files. Numerical results must therefore be generated after the corresponding datasets are inserted and the code is executed.

General Methodology

Each project follows the standard exploratory data analysis workflow: understand the problem, inspect the dataset, clean inconsistent records, summarize variables, visualize patterns, interpret findings, and provide recommendations.

EDA01 – App Insights Analysis

Project context: A technology company wants to use Google Play Store data to improve app development, user experience, downloads, and revenue.

Objectives

- Identify successful app categories.
- Study ratings, reviews, installs, size, price, and content rating.
- Compare free and paid apps.
- Identify relationships between reviews, installs, and ratings.
- Develop recommendations for app developers.

Expected dataset fields

App, Category, Rating, Reviews, Size, Installs, Type, Price, Content Rating, Genres, Last Updated, and Android Version.

Data preparation

- Remove duplicates.
- Convert Reviews and Installs to numeric values.
- Convert Size to megabytes.
- Convert Price to numeric values.
- Handle missing ratings.
- Standardize labels.

Exploratory analysis questions

- Which categories contain the most applications?
- Which categories have the highest average rating?
- Which apps have the greatest reviews and installs?
- Are free apps more widely installed than paid apps?
- Are reviews associated with ratings or installs?
- Does app size influence popularity?

Suggested visualizations

- Bar charts for categorical counts or averages.
- Histograms for numerical distributions.
- Box plots for group comparisons.
- Scatter plots for numerical relationships.
- Correlation heatmaps where appropriate.

Expected findings to report

- Popular categories may have stronger download potential.
- Review volume indicates engagement but does not guarantee high ratings.
- Free apps may reach more users, while paid apps may monetize a smaller audience.

Recommendations

- Prioritize strong-demand categories.
- Improve onboarding, stability, and support.
- Use reviews to identify complaints.
- Test pricing and advertising strategies.

Reusable Python starter code

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

df = pd.read_csv("googleplaystore.csv")
print(df.shape)
print(df.head())
print(df.info())
print(df.isnull().sum())
print(df.describe(include="all"))
df = df.drop_duplicates()
# Add project-specific cleaning and analysis below.
plt.show()
```

EDA02 – Hotel Operations Analysis

Project context: Elite Hotels International wants to understand booking patterns, guest behavior, cancellations, and operational performance.

Objectives

- Analyze booking volume and seasonal demand.
- Compare city and resort hotels.
- Measure cancellations and no-shows.
- Understand lead time, length of stay, customer type, and special requests.
- Support pricing, staffing, and room allocation.

Expected dataset fields

Hotel, Is Canceled, Lead Time, Arrival Date, Stays, Adults, Children, Country, Market Segment, Deposit Type, Customer Type, ADR, and Total Special Requests.

Data preparation

- Create an arrival date.
- Handle missing values.
- Check impossible values.
- Create total nights and guests.
- Create a cancellation indicator.
- Check duplicates.

Exploratory analysis questions

- How does booking volume vary by month?
- Which hotel type receives more bookings and revenue?
- What percentage is canceled?
- How does cancellation relate to lead time and deposit type?
- Which countries and segments generate most bookings?
- What is ADR by hotel type and season?

Suggested visualizations

- Bar charts for categorical counts or averages.
- Histograms for numerical distributions.
- Box plots for group comparisons.
- Scatter plots for numerical relationships.
- Correlation heatmaps where appropriate.

Expected findings to report

- Longer lead times may be associated with more cancellation exposure.
- Seasonal demand affects staffing, inventory, and pricing.
- Hotel type, segment, deposit type, and country reveal different behaviors.

Recommendations

- Use demand-based pricing.
- Review policies for high-risk segments.
- Improve forecasting with lead time and cancellation history.
- Allocate staff according to monthly demand.

Reusable Python starter code

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

df = pd.read_csv("hotel_bookings.csv")
print(df.shape)
print(df.head())
print(df.info())
print(df.isnull().sum())
print(df.describe(include="all"))
df = df.drop_duplicates()
# Add project-specific cleaning and analysis below.
plt.show()
```

EDA03 – Heart Health Analysis

Project context: A health dataset can be explored to identify patterns associated with the presence or absence of heart disease. Results are educational and not medical diagnosis.

Objectives

- Describe the target outcome.
- Compare demographic and clinical variables.
- Identify variables associated with heart disease.
- Create clear visual evidence.
- Discuss responsible use and limitations.

Expected dataset fields

Age, Sex, Chest Pain Type, Resting Blood Pressure, Cholesterol, Fasting Blood Sugar, Resting ECG, Maximum Heart Rate, Exercise-Induced Angina, Oldpeak, ST Slope, and Target.

Data preparation

- Inspect missing and duplicated records.
- Encode categorical variables.
- Check ranges and outliers.
- Separate predictors from target.
- Use stratified summaries.
- Do not present association as causation.

Exploratory analysis questions

- What proportion indicates heart disease?
- How does prevalence vary by age and sex?
- How do cholesterol and blood pressure differ by outcome?
- Is maximum heart rate different between groups?
- How are chest pain and exercise angina associated?
- Which variables correlate most with target?

Suggested visualizations

- Bar charts for categorical counts or averages.
- Histograms for numerical distributions.
- Box plots for group comparisons.
- Scatter plots for numerical relationships.
- Correlation heatmaps where appropriate.

Expected findings to report

- Clinical variables may show different distributions across outcome groups.
- Associations must be interpreted cautiously.
- The final report must include actual counts and percentages from the dataset.

Recommendations

- Use findings for educational awareness, not diagnosis.
- Validate findings with medical professionals.
- Protect health information.
- Document limitations and bias.

Reusable Python starter code

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

df = pd.read_csv("heart_health.csv")
print(df.shape)
print(df.head())
print(df.info())
print(df.isnull().sum())
print(df.describe(include="all"))
df = df.drop_duplicates()
# Add project-specific cleaning and analysis below.
plt.show()
```

Final Submission Checklist

- Insert each correct dataset.
- Run and save the notebook or script.
- Replace placeholder findings with computed results.
- Add chart titles and axis labels.
- Check missing-value and duplicate decisions.
- Write a conclusion for each project.
- Include sources and limitations.
- Proofread before submission.

Important: The supplied checklist names the three resources but does not contain the actual datasets. This report is a complete project framework; final numerical findings require the resource files.